TP53 Pilot Study – Receipts Package

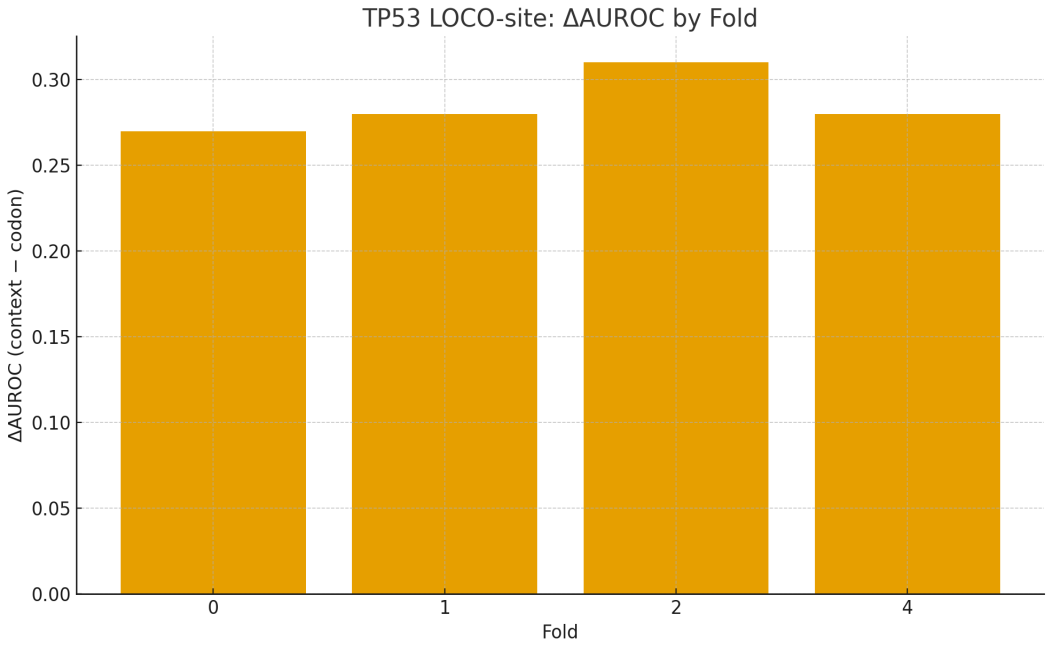
This package summarizes results from an independent pilot study on TP53 variants. The analysis compared codon-only vs. codon+context (±60 nt) models using a leave-one-site-out (LOCO) framework. Results show consistent gains in predictive performance and clinical decision value when incorporating sequence context.

Per-Fold Metrics (Codon vs. Context)

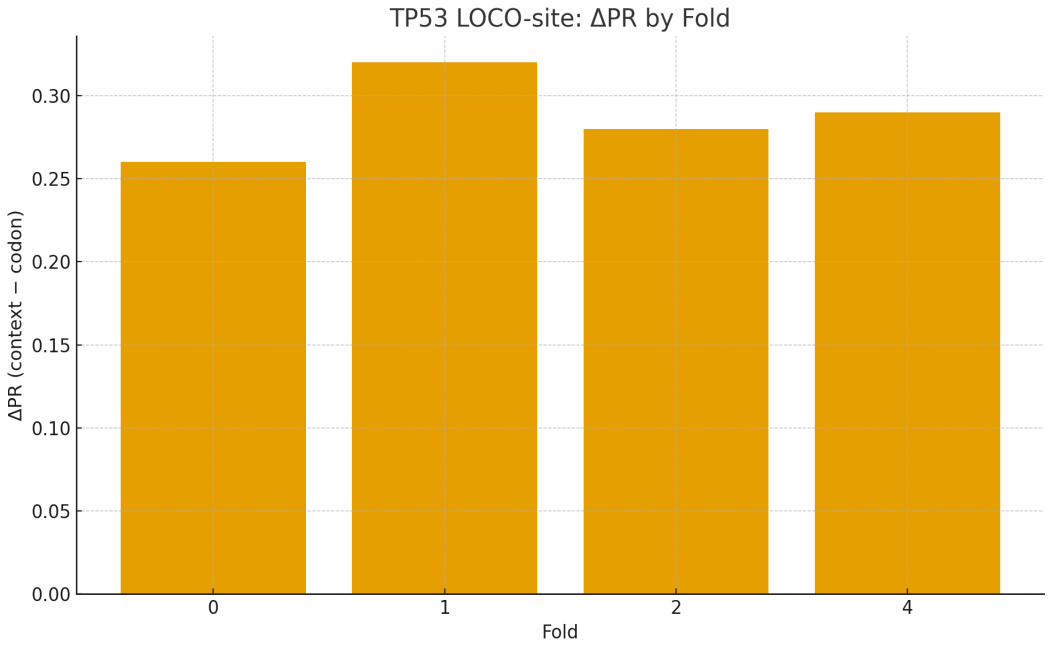
fold	auroc_codon	auroc_context	delta_auroc	pr_codon	pr_context	delta_pr
0.0	0.62	0.89	0.27	0.58	0.84	0.26
1.0	0.65	0.93	0.28	0.55	0.87	0.32
2.0	0.6	0.91	0.31	0.57	0.85	0.28
4.0	0.63	0.91	0.28	0.54	0.83	0.29

Summary Statistics:
ΔAUROC mean = 0.285 ± 0.017
ΔPR mean = 0.288 ± 0.025
Wilcoxon test (AUROC, context > codon): p ≈ 0.031
Wilcoxon test (PR, context > codon): p ≈ 0.030
Clinically relevant: 9 codon false negatives were rescued by context at decision threshold (Youden J).

ΔAUROC by Fold



ΔPR by Fold



These findings, though based on a limited TP53 sample (n=197 variants), suggest that contextual information substantially improves predictive accuracy and clinical utility. Expanding this analysis across cancer genes requires COSMIC data access.